

# Road Damage Detection & Smart Complaint Management System

## Project Overview

### 1. Problem Statement & Problem Solving

Traditional methods of identifying and repairing road damage are highly manual, time-consuming, and reactive. Citizens often face difficulties in reporting potholes, cracks, or severe road deterioration, and authorities struggle to prioritize repairs due to a lack of organized data and severity assessment.

**How this system solves the problem:** This project introduces an AI-driven, automated approach to road maintenance. Users or field agents can capture images of road damage and upload them to the system. The integrated Deep Learning model automatically detects the damage, classifies its type (e.g., Pothole, Crack), and assigns a severity level. The system then generates a standardized complaint ticket, equipped with geolocation and severity data, enabling authorities to prioritize critical repairs efficiently without manual initial inspection.

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### 2. Use Cases

- **Automated Damage Assessment:** Citizens or municipal workers upload images of damaged roads. The AI instantly evaluates the damage and creates an actionable ticket.
  - **Prioritized Ticket Resolution (Triage):** City councils and authorities can view a unified dashboard where tickets are automatically sorted or flagged based on the AI-determined severity.
  - **Smart Complaint Management:** An end-to-end ticketing system for administrators to track the lifecycle of a repair, from "Open" to "Resolved."
  - **Data-Driven Maintenance Planning:** Aggregated data over time helps governments understand which areas suffer from the most frequent damage, allowing for proactive budget allocation and better road materials research.
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### 3. Advantages

- **Actionable Intelligence:** Removes human subjectivity from severity estimation.
  - **Resource Optimization:** Helps city planners dispatch repair crews directly to the most critical locations first.
  - **Cost Efficiency:** Reduces the need for manual patrol teams to survey road conditions continually.
  - **Transparency and Engagement:** Allows citizens to actively participate in maintaining their city's infrastructure by providing an easy-to-use reporting portal.
  - **Fast Turnaround:** Automated ticket generation drastically reduces the time between identifying damage and scheduling a repair.
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### 4. Technology Stack Used

- **Frontend (User Interface & Dashboard):**
  - **React.js & Vite:** For building a fast, responsive, and dynamic user interface.
  - **CSS Modules & Recharts:** For styling and visualizing ticket statistics on the admin dashboard.
- **Backend (API & Core Logic):**
  - **Python & FastAPI:** Provides a high-performance, asynchronous REST API to handle uploads, database CRUD operations, and interface with the Machine Learning model.
  - **Pydantic & SQLAlchemy:** For robust data validation and ORM-based database interactions.

- **Automated Notifications:**
    - **Email Service (SMTP):** Used to notify users about ticket creation and status updates.
  - **Database:**
    - **SQLite:** A lightweight, file-based relational database used for storing user accounts, complaint tickets, and associated metadata.
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## 5. Algorithms & Machine Learning Models Used

The core engine of the detection system relies on state-of-the-art Computer Vision algorithms:

- **Convolutional Neural Networks (CNN):** The foundational architecture for processing and extracting features from images.
  - **MobileNetV2:** The project relies on lightweight, highly efficient object detection architectures capable of running in real-time. The model is fine-tuned specifically on a dataset of road imperfections.
  - **Image Preprocessing:** using OpenCV for resizing, normalization, and augmenting images before passing them to the inference engine.
  - **Severity Inference Logic:** Heuristics applied to the confidence score and bounding box dimensions returned by the model to classify damage as `Low` , `Medium` , or `High` severity.
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## 6. Future Improvements

- **Mobile Application Integration:** Developing dedicated iOS/Android apps to automatically capture GPS coordinates and device orientation when photographing damage.
- **Real-time Video Processing:** Mounting cameras on municipal vehicles to automatically scan and log road damage during normal driving routes.
- **Advanced Predictive Maintenance:** Using historical data to predict when continuous wear and tear on a specific road segment will turn into a dangerous pothole.
- **GIS (Geographic Information System) Integration:** Visualizing tickets on a live map interface, clustering damage locations to optimize the routes taken by repair crews.
- **Automated Escalation:** Implementing a cron-job or background worker to automatically escalate tickets that remain unresolved beyond their SLA (Service Level Agreement).